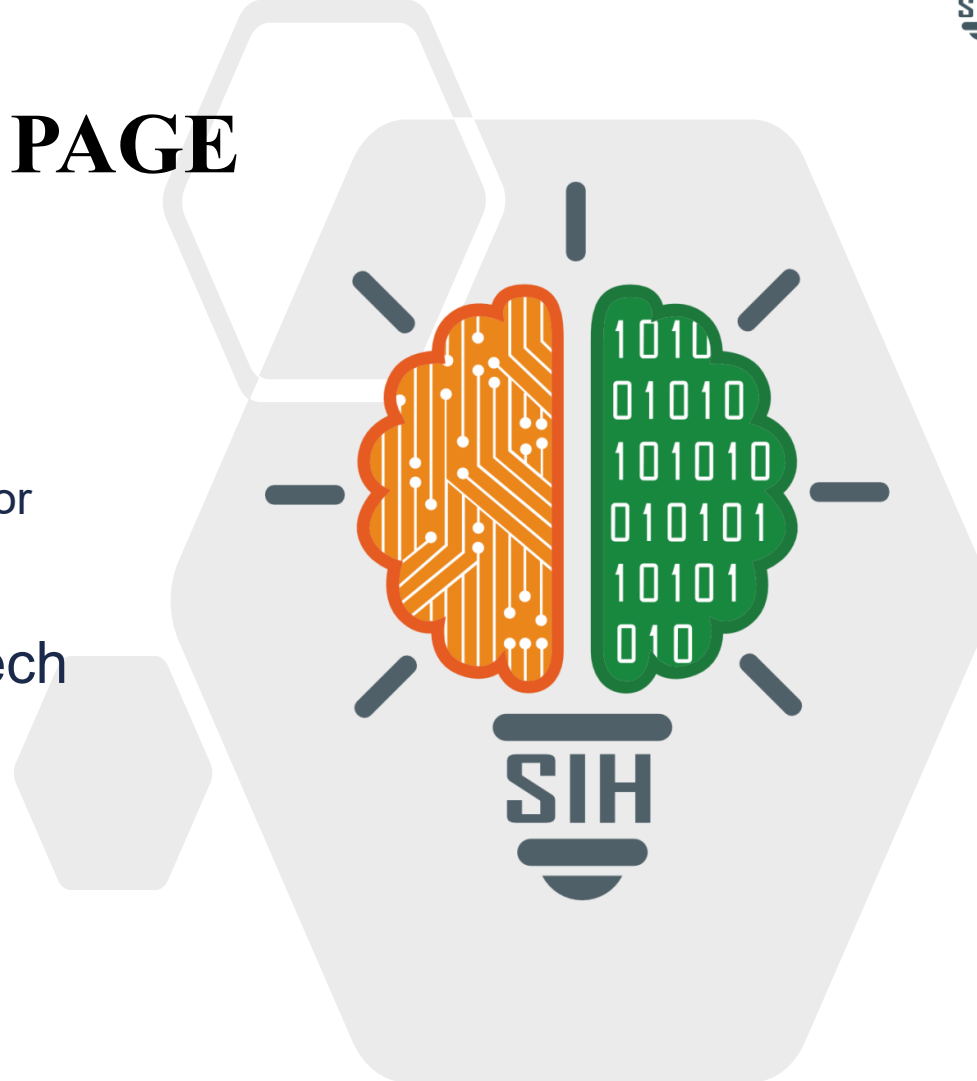


TITLE PAGE

- **Problem Statement ID** – SIH26186
- **Problem Statement Title** – AI-Based Predictive Personnel Stress and Welfare Monitoring System for Uniformed Forces
- **Theme** – MedTech / BioTech / HealthTech
- **PS Category** – Software
- **Team ID** – _____
- **Team Name** – Team Name



MANOBAL – AI Welfare Intelligence

Predictive Stress & Burnout Monitoring for Uniformed Forces • welfare-first • privacy-first • explainable



Proposed Solution

MANOBAL (मनोबल = morale) is a privacy-first AI platform that fuses HRMS signals (leave patterns, deployment history, duty hours, transfer frequency), voluntary self-assessments (PHQ-9 / GAD-7 / PSS-10 + 2-minute mood check-ins) and opt-in wearable data into one explainable Stress & Burnout Risk Index — and converts it into welfare actions, never disciplinary flags.



How It Addresses the Problem

- Replaces manual observation & self-reporting with continuous early-warning analytics — rising risk trends are flagged weeks before burnout, not after an incident
- Welfare Officer gets a ranked, explained caseload; Commander sees only unit-level heatmaps (no names) — dignity & confidentiality preserved
- Auto-recommends interventions: counselling (Tele-MANAS referral), leave prioritisation, workload rebalancing, buddy pairing, family outreach



Innovation & Uniqueness

- Consent-tiered data (Tier 0: HRMS only → Tier 1: self-report → Tier 2: wearables) — every jawan controls what is shared
- Explainable AI (SHAP) shows “why flagged” in Hindi/English; human-in-the-loop — AI ranks, the welfare officer decides
- Stigma-safe by design: pseudonymised IDs, k-anonymity, welfare-only access, never linked to ACR / promotion / posting
- Offline-first app in 12 Indian languages for LWE / border / high-altitude postings; force-specific model calibration

MANOBAL AT A GLANCE

SIGNALS (consent-tiered, opt-in)



HRMS Signals
leave · duty hrs · transfers



Self-Report App
PHQ-9 · GAD-7 · mood



Wearables (opt-in)
HRV · sleep · activity



Predictive Behavioural Engine

Explainable Stress & Burnout Risk Index (0–100) + top-3 drivers + recommended intervention



Welfare Officer
ranked & explained caseload;
alerts; case notes

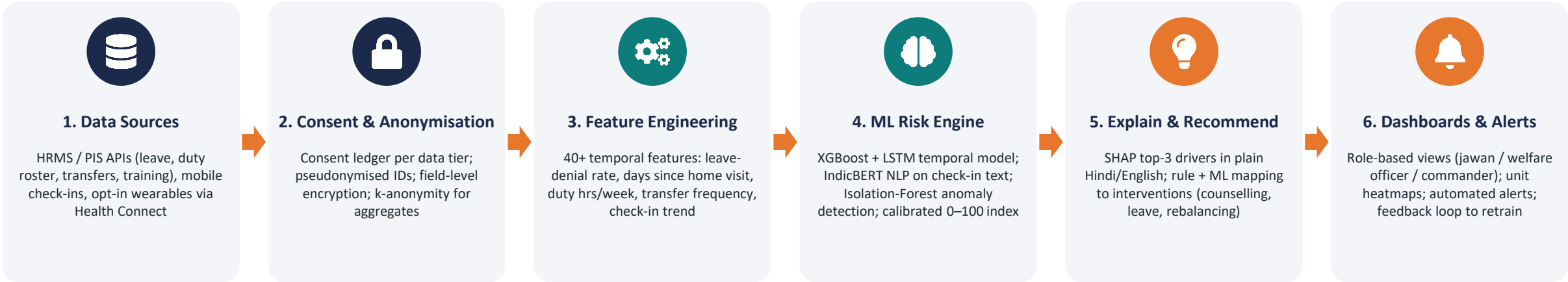


Commander / HQ
unit-level heatmaps & trends
only — no names



Privacy Shield

pseudonymised IDs · k-anonymity · AES-256 · DPDP Act 2023 · welfare-only RBAC · full audit trail



TECHNOLOGY STACK

**Mobile App**
Flutter (Android/iOS), offline-first SQLite sync, Hindi + 12 Indian languages, voice check-ins

**Backend & APIs**
Python FastAPI + Node.js, REST/gRPC, Keycloak RBAC & MFA, HRMS connectors

**AI / ML**
scikit-learn, XGBoost, PyTorch (LSTM), SHAP, Hugging Face IndicBERT, MLflow

**Data Layer**
PostgreSQL, Redis, Feast feature store, MinIO object store, Apache Airflow pipelines

**Security & Privacy**
AES-256 at rest, TLS 1.3, HashiCorp Vault, immutable audit logs, DPDP Act 2023 controls

**Deployment**
Docker + Kubernetes on NIC MeghRaj / on-prem, air-gap ready, CERT-In hardening

METHODOLOGY & IMPLEMENTATION PROCESS

1 Integrate: Secure HRMS/PIS connectors; build synthetic + anonymised pilot dataset with force HR & medical wing

2 Co-design: Assessments & thresholds with force psychologists; validated scales (PHQ-9, GAD-7, PSS-10)

3 Train & validate: Time-series models; optimise for recall (missed cases cost most); fairness & drift checks

4 Pilot (90 days): One battalion (~1,000 personnel); measure lead-time, uptake, counsellor workload, false-positive rate

5 Scale & learn: Force-wide rollout via MeghRaj; continuous learning from welfare-officer feedback

✓ Feasibility

✓ Software-only — no new hardware; wearables strictly optional

✓ Leave, duty & transfer data already digitised in CAPF PIS / e-HRMS

✓ Validated clinical scales (PHQ-9, GAD-7, PSS-10) — no new instruments to prove

✓ Open-source ML stack; XGBoost / LSTM / SHAP are mature & auditable

✓ Aligned with MHA Task Force (2022) mandate on stress & suicide prevention

⚠ Challenges & Risks

! Trust & stigma — fear that a risk flag will affect career or posting

! Cyber threats on highly sensitive psychological & welfare data

! False positives / negatives; cold-start with little labelled data

! Low self-reporting in remote, low-connectivity deployments

! Cultural & linguistic diversity across forces and units

🔧 Mitigation Strategies

➔ Welfare-only charter: no link to ACR/promotion; jawan can view & contest own score

➔ Pseudonymisation, AES-256, RBAC, on-prem/MeghRaj hosting, audit trails, DPDP-compliant

➔ Human-in-the-loop; thresholds tuned for recall; feedback loop retrains monthly

➔ Offline-first app, 2-min check-ins, 12 languages, voice input

➔ Force-specific calibration with unit psychologists; bias audits each quarter

₹ ₹0 hardware

Software-only platform; opt-in wearables use devices personnel already own

⚡ Zero licence cost

100 % open-source stack; indigenous IP retained by MHA / forces

☁ Existing infra

Runs on NIC MeghRaj or on-prem; air-gap ready for sensitive units

🔒 Opt-in & DPDP-ready

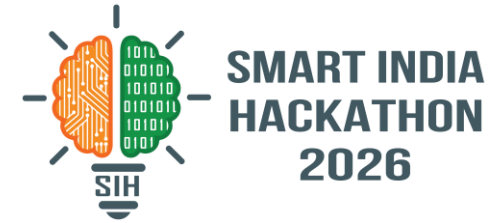
Consent-based tiers — no new legal mandate needed to start a pilot

PHASED ROLLOUT ROADMAP



Team
Name

IMPACT AND BENEFITS



730

suicides in CAPFs, NSG & Assam Rifles in 2020–24 (MHA, Rajya Sabha, Dec 2024)

55,500+

voluntary retirements & resignations in the same period — stress and attrition are linked

Reactive → Proactive

MANOBAL moves welfare from post-incident response to early, evidence-based prevention

IMPACT

Early identification — of personnel needing welfare support — timely counselling & leave



Fewer incidents — reduction in stress-related incidents, self-harm and fratricide



Balanced workload — data-driven duty rosters, leave planning & transfer decisions



Higher retention — lower attrition / VRS; better job satisfaction & morale



Evidence-based welfare — unit-level trends guide budgets, counsellor deployment & policy



BENEFITS



Social — confidential, destigmatised mental-health support for jawans & families



Organisational — enhanced force readiness, resilience & operational effectiveness



Economic — saves recruitment & training cost of every trained personnel retained



Strategic — indigenous, sovereign welfare-tech for Indian forces (Atmanirbhar Bharat)



Scalable — State Police, NDRF / disaster response, government & corporate HR wellness



Policy & Government Data

1. MHA reply, Rajya Sabha (4 Dec 2024): 730 suicides, 47,891 VRS & 7,664 resignations in CAPFs/NSG/AR, 2020–24
<https://aninews.in/news/national/general-news/730-suicides-amongst-capfs-nsg-and-assam-rifles-during-the-last-five-years-nityanand-rai20241204183618/>
2. MHA Task Force on suicides & fratricide in CAPFs (2022) — service, working & personal stress factors
<https://www.deccanherald.com/india/mha-creates-taskforce-to-prevent-suicides-by-central-armed-police-forces-troops-1056571.html>
3. Digital Personal Data Protection Act, 2023 — MeitY (consent, purpose limitation, data-fiduciary duties)
<https://www.meity.gov.in/data-protection-framework>



Clinical Instruments & Existing Welfare Measures

4. Kroenke et al. (2001) PHQ-9 depression scale; Spitzer et al. (2006) GAD-7 anxiety scale — validated screening tools
<https://pubmed.ncbi.nlm.nih.gov/11556941/>
5. Cohen et al. (1983) Perceived Stress Scale (PSS-10); WHO ICD-11 burn-out definition (2019)
<https://www.who.int/news/item/28-05-2019-burn-out-an-occupational-phenomenon-international-classification-of-diseases>
6. Tele-MANAS (MoHFW) national tele-mental-health helpline 14416 — counselling referral pathway; CRPF ‘Chaupal’ & buddy system
<https://telemanas.mohfw.gov.in/>



Research

7. Schmidt et al. (2018) WESAD — wearable stress & affect detection dataset (ICMI)
<https://dl.acm.org/doi/10.1145/3242969.3242985>
8. Lundberg & Lee (2017) SHAP — unified approach to interpreting model predictions (NeurIPS)
<https://arxiv.org/abs/1705.07874>
9. Chen & Guestrin (2016) XGBoost — scalable tree boosting system (KDD)
<https://arxiv.org/abs/1603.02754>



Technology & Standards

10. Hugging Face IndicBERT / AI4Bharat — Indian-language NLP for check-in text
<https://huggingface.co/ai4bharat>
11. NIC MeghRaj Government Cloud & CERT-In security guidelines for hosting sensitive data
<https://cloud.gov.in/>
12. Android Health Connect API — consent-based access to wearable HRV, sleep & activity data
<https://developer.android.com/health-and-fitness/guides/health-connect>